

Figure Trace Guide

How every figure in `results/figures/` was generated, what data it uses, and where it feeds into the project's analysis pipeline.

Colour-Tier Definitions (Precinct Demand)

Several figures classify Manhattan's 25 precincts into three demand tiers:

Tier	Colour	Rule	Meaning
High	Red (<code>#C0392B</code>)	crashes/day $\geq$ Q75 (4.8)	Top 25 % of precincts
Medium	Blue (<code>#5B8DBE</code>)	Q25 $\leq$ crashes/day < Q75	Middle 50 %
Low	Green (<code>#27AE60</code>)	crashes/day < Q25 (2.0)	Bottom 25 %

Thresholds are derived from the 25th and 75th percentiles of the `crashes_per_day` column computed in `data/processed/demand_lambda_precinct.csv` (hourly rate x 24).

Figures by Category

Demand & EDA Figures

#	File	What it shows	Data source	Generated by	Down-stream usage
1	<code>fig_hourly_demand.png</code>	24-hour crash demand profile	<code>crashes_manhattan.csv</code>	<code>notebooks/02_eda_spatiotemporal.ipynb</code>	Calibrates NHPP hourly factors in <code>de-mand_lambda_hourly.csv</code> ; informs time-of-day speed factors in <code>configs/service.yaml</code>
2	<code>fig_daily_demand.png</code>	Day-of-week crash volume	<code>crashes_manhattan.csv</code>	<code>notebooks/02_eda_spatiotemporal.ipynb</code>	Calibrates <code>de-mand_lambda_dow.csv</code>
3	<code>fig_temporal_trends.png</code>	Long-term crash trends 2012-2026	<code>crashes_manhattan.csv</code>	<code>notebooks/02_eda_spatiotemporal.ipynb</code>	Supports Assumption A1 (stationarity); discussed in technical report Section 5.1
4	<code>fig_crash_heatmap.png</code>	Spatial heatmap of crash incidents	<code>crashes_manhattan.csv</code> , precinct GeoJSON	<code>notebooks/02_eda_spatiotemporal.ipynb</code>	Motivates spatial optimization; identifies hotspots for P2 allocation
5	<code>fig_precinct_density.png</code>	Precinct demand density choropleth	<code>crashes_manhattan.csv</code> , precinct boundaries	<code>notebooks/02_eda_spatiotemporal.ipynb</code>	Companion to <code>fig_precinct_demand.png</code> ; normalises by precinct area
6	<code>fig_precinct_demand.png</code>	Original precinct demand bar chart	<code>de-mand_lambda_precinct.csv</code>	<code>scripts/demand_modeling.py</code>	Referenced in technical report Table 17;

#	File	What it shows	Data source	Generated by	Down-stream usage
		(red/blue/green)			motivates demand-weighted optimization
7	<code>pre-cinct_demand_rates_improved.png</code>	Improved bar chart with explicit legend, Q25/Q75 annotations, value labels	<code>de-mand_lambda_precinct.csv</code>	<code>scripts/generate_precinct_demand_visualizations.py</code>	Replaces Fig 6 where a clearer legend is needed
8	<code>pre-cinct_demand_heatmap.png</code>	Spatial choropleth: precinct polygons coloured by demand rate	<code>de-mand_lambda_precinct.csv</code> , <code>Po-lice_Precincts_20260223.csv</code> , <code>manhattan_boundary.geojson</code> , <code>cbd_boundary.geojson</code>	<code>scripts/generate_precinct_demand_visualizations.py</code>	Supplements the bar chart with geographic context; links demand distribution to firehouse locations
9	<code>fig_demand_model_fit.png</code>	NHPP model fit diagnostics (observed vs predicted)	<code>de-mand_lambda_*.csv</code> , <code>crashes_manhattan.csv</code>	<code>scripts/demand_modeling.py</code>	Validates the NHPP arrival model used in simulation
10	<code>fig_hourly_rates.png</code>	CBD vs non-CBD hourly and day-of-week lambda factors	<code>crashes_manhattan.csv</code> , <code>cbd_boundary.geojson</code>	<code>scripts/demand_modeling.py</code>	Confirms CBD temporal pattern similarity; informs CBD analysis

Firehouse & Infrastructure Figures

#	File	What it shows	Data source	Generated by	Down-stream usage
11	<code>fig_firehouses_map.png</code>	Map of 48 Manhattan FDNY fire-houses	<code>fire-houses_manhattan.csv</code>	<code>notebooks/02_eda_spatiotemporal.ipynb</code>	Documents candidate staging locations for all optimization models
12	<code>distance_matrix_heatmap.png</code>	Haversine distance matrix (48 fire-houses x 30 precincts)	<code>distance_matrix_fire-house_precinct.csv</code>	<code>notebooks/04_service_travel_proxy.ipynb</code>	Validates distance computation; input to travel-time proxy

Service & Travel Model Figures

#	File	What it shows	Data source	Generated by	Down-stream usage
13	<code>travel_time_by_tod.png</code>	Travel time distributions by time-of-day band	Distance matrix, <code>configs/service.yaml</code>	<code>notebooks/04_service_travel_proxy.ipynb</code>	Validates time-of-day speed factors
14	<code>tod_speed_factors.png</code>	24-hour speed factor profile	<code>configs/service.yaml</code>	<code>notebooks/04_service_travel_proxy.ipynb</code>	Documents the speed multiplier curve
15	<code>service_time_distribution.png</code>	LogNormal service time distribution	<code>configs/service.yaml</code> parameters	<code>notebooks/04_service_travel_proxy.ipynb</code>	Validates service time model; referenced in service model spec
16	<code>nhpp_arrivals_demo.png</code>	NHPP thinning algorithm demonstration	<code>demand_lambda_*.csv</code>	<code>notebooks/04_service_travel_proxy.ipynb</code>	Validates arrival generation used in simulation engine

Optimization Figures

#	File	What it shows	Data source	Generated by	Down-stream usage
17	<code>fig_policy_comparison.png</code>	Summary policy comparison (P0, P1, P2 variants)	Optimization solver output	<code>scripts/run_optimization_comparison.py</code>	Key result figure; drives policy recommendation
18	<code>fig_tradeoff_curve.png</code>	Response time vs coverage trade-off across fleet sizes (original, has overlapping labels)	Optimization solver output	<code>scripts/run_optimization_comparison.py</code>	Informs fleet sizing decision (K=20-30). See <code>docs/policy_tradeoff_analysis.md</code> for full interpretation. P0 = uniform baseline (not spatially-stratified).
18a	<code>re-sponse_time_coverage_tradeoff_improved.png</code>	Improved trade-off plot with jittered points, marker shapes per policy, and side-by-side data table	<code>policy_comparison.csv</code>	<code>scripts/generate_tradeoff_improved.py</code>	Replaces fig 18 for reports; see <code>docs/policy_tradeoff_analysis.md</code>
18b	<code>re-sponse_time_coverage_tradeoff_zoomed.png</code>	Zoomed view of high-performance region (RT < 5 min) showing all overlapping points separated	<code>policy_comparison.csv</code>	<code>scripts/generate_tradeoff_improved.py</code>	Detail view for technical report appendix
19	<code>opt_allocation_comparison.png</code>	Allocation map comparison across models	Solver allocations	<code>scripts/run_optimization_comparison.py</code>	Shows spatial distribution of units by policy

#	File	What it shows	Data source	Generated by	Down-stream usage
20	opt_inputs.png	Optimization inputs (travel times, demand weights)	Distance matrix, demand data	scripts/run_optimization_comparison.py	Verifies input correctness
21	opt_sensitivity.png	Objective value vs fleet size	Solver output	scripts/run_optimization_comparison.py	Supports diminishing-returns analysis

Simulation & Experiment Figures

#	File	What it shows	Data source	Generated by	Down-stream usage
22	exp1_policy_comparison.png	Experiment 1: P0 vs P1 vs P2 box plots	Simulation replications	notebooks/07_production_results.ipynb	Primary evidence for P2 dominance; technical report Section 5.5
23	exp2_fleet_sensitivity.png	Experiment 2: Response time vs fleet size	Simulation replications	notebooks/07_production_results.ipynb	Fleet sizing recommendation; technical report Section 5.6
24	exp3_demand_sensitivity.png	Experiment 3: Response time vs demand multiplier	Simulation replications	notebooks/07_production_results.ipynb	Demand robustness analysis; technical report Section 5.7
25	exp4_service_robustness.png	Experiment 4: Response time vs service time mean	Simulation replications	notebooks/07_production_results.ipynb	Service time sensitivity; technical report Section 5.7

Verification & Validation Figures

#	File	What it shows	Data source	Generated by	Down-stream usage
26	verification_toy_timeline.png	Toy example event timeline trace	Synthetic toy scenario	notebooks/06_simulation_debug.py	V&V evidence; confirms simulation mechanics
27	validation_p0_vs_p2.png	Pilot 1: P0 vs P2 directional comparison	Validation simulation runs	notebooks/06_simulation_debug.py	V&V evidence; confirms P2 dominance direction
28	validation_sensitivity_K.png	Pilot 2: Response time vs fleet size K	Validation simulation runs	notebooks/06_simulation_debug.py	V&V evidence; confirms monotonic decrease
29	validation_sensitivity_demand.png	Pilot 3: Response time vs demand intensity	Validation simulation runs	notebooks/06_simulation_debug.py	V&V evidence; confirms monotonic increase

Production V2 & Fleet Analysis Figures

#	File	What it shows	Data source	Generated by	Down-stream usage
30	fleet_sensitivity_v2_dual.png	Fleet sensitivity dual-panel (mean RT and coverage)	Production V2 simulation	scripts/run_production_v2.py	Technical report Section 5.6 key figure
31	response_time_distribution_by_policy.png	RT distribution violin/box by policy and K	Production simulation	Production analysis pipeline	Technical report Section 5.5
32	policy_comparison_panel_K20_cap2.png	Three-panel staging map: P0, P1, P2 at K=20 cap=2	Allocation results	Production analysis pipeline	Technical report Section 5.4 map figure

CBD Analysis Figures

#	File	What it shows	Data source	Generated by	Down-stream usage
33	<code>cbd_heatmap.png</code>	CBD area crash density and firehouse locations	<code>crashes_manhattan.csv</code> , <code>cbd_boundary.geojson</code>	<code>notebooks/09_cbd_analyses.ipynb</code>	Contextualises CBD-focused analysis
34	<code>cbd_response_comparison.png</code>	RT comparison between policies under CBD stress	CBD simulation runs	<code>notebooks/09_cbd_analyses.ipynb</code>	Technical report Section 5.10
35	<code>cbd_scenario_comparison.png</code>	CBD vs Manhattan-wide scenario comparison	CBD experiment results	<code>notebooks/09_cbd_analyses.ipynb</code>	Technical report Section 5.10
36	<code>cbd_robustness_enhanced.png</code>	Enhanced CBD robustness (RT and coverage)	CBD experiment	Production analysis pipeline	Technical report Section 5.10
37	<code>cbd_equity_tradeoff_summary.png</code>	CBD equity-efficiency trade-off	CBD-focused optimisation	<code>scripts/run_cbd_focused_optimization.py</code>	Supports DEC-009 (CBD-focused models not recommended)
38	<code>fig_cbd_comparison.png</code>	CBD-focused robustness comparison	<code>crashes_manhattan.csv</code> , CBD boundary	<code>notebooks/02_eda_spatiotemporal.ipynb</code>	EDA-level CBD context

Queue & Seasonal Analysis Figures

#	File	What it shows	Data source	Generated by	Down-stream usage
39	queue_comparison_by_policy.png	Queue metrics by policy	Simulation queue logs	scripts/analyze_queue_metrics.py	Technical report Section 5.8
40	queue_heatmap.png	Queue length heatmap	Simulation queue logs	scripts/analyze_queue_metrics.py	Congestion analysis
41	queue_vs_demand.png	Queue metrics vs demand multiplier	Simulation queue logs	scripts/analyze_queue_metrics.py	Capacity planning
42	queue_vs_fleet_size.png	Queue metrics vs fleet size	Simulation queue logs	scripts/analyze_queue_metrics.py	Fleet sizing
43	seasonal_patterns.png	Seasonal variation analysis	crashes_manhattan.csv	scripts/analyze_seasonal_patterns.py	Technical report Section 5.9
44	seasonal_decomposition.png	Seasonal decomposition of monthly demand	crashes_manhattan.csv	scripts/analyze_seasonal_patterns.py	Supports Assumption A1
45	seasonal_heatmap.png	Month x day-of-week crash heatmap	crashes_manhattan.csv	scripts/analyze_seasonal_patterns.py	Temporal pattern documentation

Distance Metric Comparison Figures

#	File	What it shows	Data source	Generated by	Down-stream usage
46	dis-distance_comparison/dis-distance_matrices_heatmap.png	Haversine vs Manhattan distance heatmaps	Distance matrices	scripts/run_distance_comparison_experiment.py	Technical report Section 5.10
47	dis-distance_comparison/dis-distance_scatter.png	Haversine vs Manhattan scatter	Distance matrices	scripts/run_distance_comparison_experiment.py	Validates distance metric robustness
48	dis-distance_comparison/dis-distance_comparison_bar.png	Performance comparison bar chart	Simulation results	scripts/run_distance_comparison_experiment.py	Key evidence for DEC-008
49	dis-distance_comparison/dis-distance_comparison_boxplot.png	Replication distribution box plots	Simulation results	scripts/run_distance_comparison_experiment.py	Statistical comparison

Capacity Sensitivity Figures

#	File	What it shows	Data source	Generated by	Down-stream usage
50	capa-city_sensitivity_heatmap.png	Mean RT by policy and capacity limit	Capacity experiment	scripts/generate_capacity_sensitivity_heatmap.py	Supports DEC-010 (cap=2 recommendation)
51	fire-house_capacity_analysis.png	Capacity analysis detail	Capacity experiment	scripts/capacity_sensitivity_analysis.py	Detailed capacity analysis

Spatial Baseline (P0) Figures

#	File	What it shows	Data source	Generated by	Down-stream usage
52	p0_spatial_map.png	P0-spatial firehouse selections on map	fire-houses_manhattan.csv	scripts/p0_spatial_analysis.py	Supports DEC-011 (spatially-stratified P0)
53	p0_spatial_metrics.png	Stratification method comparison metrics	fire-houses_manhattan.csv	scripts/p0_spatial_analysis.py	Method selection evidence
54	p0_spatial_north_south.png	North-south distribution analysis	fire-houses_manhattan.csv	scripts/p0_spatial_analysis.py	Spatial fairness documentation

Publication-Quality Figures

#	File	What it shows	Data source	Generated by	Down-stream usage
55	pub_fig1_policy_comparison.png	Publication: Policy comparison	Production results	scripts/generate_publication_figure_s.py	Final report Figure 1
56	pub_fig2_fleet_sensitivity.png	Publication: Fleet sensitivity	Production results	scripts/generate_publication_figure_s.py	Final report Figure 2
57	pub_fig3_demand_and_robustness.png	Publication: Demand robustness	Production results	scripts/generate_publication_figure_s.py	Final report Figure 3
58	pub_fig4_service_sensitivity.png	Publication: Service time sensitivity	Production results	scripts/generate_publication_figure_s.py	Final report Figure 4
59	pub_fig5_performance_heatmap.png	Publication: Performance heatmap	Production results	scripts/generate_publication_figure_s.py	Final report Figure 5
60	project_summary_dashboard.png	Full project summary dashboard	All key results	scripts/generate_summary_dashboard.py	Executive overview

- 3. **Policy Evaluation:** Comparing P0 (uniform) with P2 (demand-weighted) reveals how much response time improves when allocation follows the demand distribution shown in these figures.
- 4. **CBD Analysis:** The heatmap’s CBD boundary overlay shows that high-demand precincts cluster in Midtown (Pcts 14, 18, 19), supporting the decision to analyse CBD-specific scenarios separately.

Cross-Reference to Technical Report Sections

Figure	Tech Report Section	Purpose
fig_hourly_demand.png	Section 5.1 (EDA)	Temporal pattern evidence
fig_precinct_demand.png	Section 5.2 (Demand Model)	Spatial demand distribution
pre-precinct_demand_rates_improved.png	Section 5.2 (improved version)	Clearer demand tier communication
precinct_demand_heatmap.png	Section 5.2 (supplement)	Geographic context
exp1_policy_comparison.png	Section 5.5 (Policy Comparison)	Primary result
exp2_fleet_sensitivity.png	Section 5.6 (Fleet Sensitivity)	Fleet sizing
exp3_demand_sensitivity.png	Section 5.7 (Robustness)	Demand stress
cbd_scenario_comparison.png	Section 5.10 (CBD)	CBD robustness
capacity_sensitivity_heatmap.png	Section 5.12 (Capacity)	Capacity constraint
validation_p0_vs_p2.png	Section 4.4.3 (V&V)	Validation pilot 1

Regeneration Commands

```
# Precinct demand visualisations (bar chart + heatmap)
python scripts/generate_precinct_demand_visualizations.py

# Demand model figures
python scripts/demand_modeling.py

# Publication-quality figures
python scripts/generate_publication_figures.py

# Queue analysis figures
python scripts/analyze_queue_metrics.py

# Seasonal pattern figures
python scripts/analyze_seasonal_patterns.py

# Capacity sensitivity heatmap
python scripts/generate_capacity_sensitivity_heatmap.py

# Summary dashboard
python scripts/generate_summary_dashboard.py

# P0 spatial analysis
python scripts/p0_spatial_analysis.py
```